AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2024

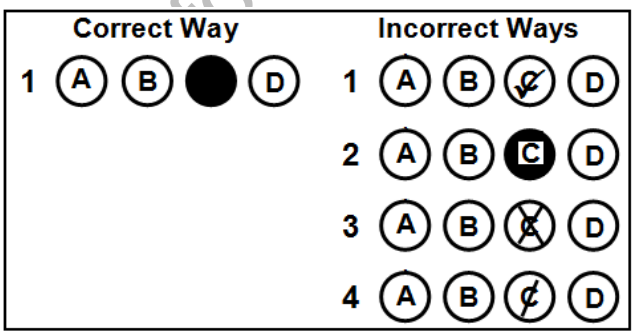
Computer Science Paper I

Time: 1 hour 30 minutes Marks: 50

AKU-EB Annual Examinations 2024 for Teaching & Learning only

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. An example of single-user operating system is

   A. DOS.
   B. UNIX.
   C. Linux.
   D. MacOS.

2. There are three processes J, K and L ready for execution. Each process will require some specific CPU time for its execution.

   CPU Time for Process J = 6 milliseconds
   CPU Time for Process K = 3 milliseconds
   CPU Time for Process L = 8 milliseconds

   What should be the sequence of execution of these processes so that average waiting time for these processes becomes minimum?

   A. J K L
   B. J L K
   C. K J L
   D. L K J

3. Consider the following statements.

   I. It is a sequence of instruction within a process.
   II. It can be executed independently.
   III. It is an instance of a program.

   The statement(s) which is/ are CORRECT about threads in a process is/ are

   A. I only.
   B. III only.
   C. I and II.
   D. II and III.

4. If an unexpected keyboard malfunction occurs, then the function of operating system to deal such event is

   A. command-interpreter.
   B. process management.
   C. interrupt handling.
   D. protection system.

5. A process has requested a resource, but that resource is currently unavailable.

   The state in which the process enters while it waits for the requested resource to become available is

   A. new.
   B. blocked.
   C. running.
   D. terminated.

Annual U-EB ations 2024 for Teachi arning only

6. The type of an operating system that is used by ATM is

   A. multiprocessor operating system.
   B. multiprogramming batch system.
   C. time-sharing operating system.
   D. real-time operating system.

7. The task that is NOT performed in the development phase of SDLC is

   A. design the database.
   B. design the flowchart and algorithms.
   C. generate the code for dataflow process.
   D. generate the code for actual user interface.

8. Consider the given responsibilities.

   I. Write programs to solve problems
   II. Analyse the mistakes in the program
   III. Analyse the data processing requirements
   IV. Investigate the problem and plan the solution

   The responsibilities of a system analyst are

   A. I and II.
   B. I and IV.
   C. II and III.
   D. III and IV.

9. The compiler shows error while executing the given variable name.

   $pecial,T@sk

   The compiler will execute this variable name if it is written as

   A. $pecial_T@sk
   B. Special T@sk
   C. Special_Task
   D. $pecial,Task

10. The escape sequence which is used to produce beep sound is

    A. \a
    B. \b
    C. \r
    D. \t

11. Which of the following expressions has NO relational operators?

    A. (x!=y)||(y!=z)
    B. (x!=y)||!(y&&z)
    C. (x>=y)&&(y&&z)
    D. (!x||y)&&(!y&&z)

12. Consider the given value.

-32,765

The integer type of a variable that CANNOT store this value is

A. int
B. long int
C. short int
D. unsigned short int

Use the given program to answer Q.13 and Q.14.

```
#include <iostream>
using namespace std;
int a, b;
int main ()
{
  int j = 2, k = 3;
  j == 5;
  k = 8;
  a = k + j;
  b = j - k;
  cout <<a <<"\n";
  cout << -b;
  return 0;
}
```

AKU-EB Annual Examinations 2024 Teaching & Learning only

13. The number of times arithmetic operators are used in this program is

A. two.
B. three.
C. four.
D. five.

14. The number of variables initialised in this program is

A. two.
B. four.
C. six.
D. eight.

15. Consider the given statement.

float L = 1.9838474;

The value of L will be displayed in output as

A. 1.9e+06
B. 1.98385
C. 1.983e+04
D. 1.98384e+02

16. Consider the following header files.

I. iostream.h
II. conio.h
III. string.h
IV. iomanip.h

The header files that are required to use cin, strcmp and setprecision in a program are

A. I and II.
B. III and IV.
C. I, II and IV.
D. I, III and IV.

17. Consider the following operators.

I. ||
II. !
III. /=
IV. +

The CORRECT arrangement of types of operators on the basis of the precedence is

	Low Precedence ⟶ **High Precedence**			
A	/=	||	!	+
B	!	+	||	/=
C	+	/=	!	||
D	||	!	/=	+

18. Consider the given program.

```
#include <iostream>
using namespace std;
int main()
{
  int X;
  double Y = 4.99;
  X = Y;
  cout << "X = " << X << endl;
  cout << "Y = " << Y << endl;
  return 0;
}
```

The number of times a manipulator is used in this program is

A. one.
B. two.
C. three.
D. four.

19. Consider the given program.

```
#include <iostream>
using namespace std;
int main ()
{
  int w=20;
  do
  {
    w=w+1;
    cout<<w<<endl;
  }  while(w<15 );
  return 0;
}
```

Upon execution, the program will display

A. 14
B. 15
C. 20
D. 21

AKTU-EB Annual Examinations 2024 for Teaching & Learning Only

20. Consider the given program.

```
#include <iostream>
using namespace std;
int main()
{
  int x;
  for(int i=1; i<=3; i++)
  {
  x=i*2;
  }
  cout<<x;
  return 0;
}
```

The program can be rewritten by using while loop as

`#include <iostream>`
`using namespace std;`
`int main()`
`{`
`  int x, i=1;`
`  while(i<3)`
`  {`
`  x=i*2;`
`  i=i+1;`
`  }`
`  cout<<x;`
`  return 0;`
`}`
	`#include <iostream>`
	`using namespace std;`
	`int main()`
	`{`
	`  int x, i=1;`
	`  while(i<=3)`
	`  {`
	`  x=i*2;`
	`  i=i+1;`
	`  }`
	`  cout<<x;`
	`  return 0;`
	`}`

A	B

`#include <iostream>`
`using namespace std;`
`int main()`
`{`
`  int x, i=1;`
`  while (i<3)`
`  {`
`  x=i*2;`
`  }`
`  cout<<x;`
`  return 0;`
`}`
	`#include <iostream>`
	`using namespace std;`
	`int main()`
	`{`
	`  int x, i=1;`
	`  while (i<=3)`
	`  {`
	`  x=i*2;`
	`  }`
	`  cout<<x;`
	`  return 0;`
	`}`

C	D

AKU-EB Annual Examinations 2024 for Teaching & Learning only

21. Which of the following gives back the control of flow of execution to the calling function?

A. exit
B. return
C. break
D. default

22. Consider the given syntax.

```
if (condition)
{
        //statement
}
else
{
        //statement
}
```

The structure identified from this syntax is

A. if-else statement.
B. else-if statement.
C. nested if statement.
D. simple if statement.

23. The programming structure that can calculate the sum of first ten even numbers is

A. loop.
B. array.
C. string.
D. selection.

Annual Exa 3 s 2024
for Teaching only

24. Consider the given program.

```
#include <iostream>
using namespace std;
int main ()
{
   char ch='q';
   switch(ch)
   {
      case 'P' :
         cout<<"Computer"<<endl;
         break;
      case 'Q' :
         cout<<"Mathematics"<<endl;
         break;
      case 'R' :
         cout<<"Physics"<<endl;
         break;
      default :
         cout<<"English"<<endl;
         break;
   }
return 0;
}
```

The output of this program is

A. Physics
B. English
C. Computer
D. Mathematics

25. Consider the given program.

```
#include <iostream>
using namespace std;
int main()
{
   for (int n = 0; n < 10; n = n + 2)
   {
   cout << n << " ";
   }
return 0;
}
```

The output of this program is

A. 1 3 5 7 9
B. 0 2 4 6 8
C. 0 2 4 6 8 10
D. 0 1 2 3 4 5 6 7 8 9

AKU-EB Annual Examinations 2024 for Teaching & Learning only

26. Consider the given program.

```
#include <iostream>
using namespace std;
int main ()
{
   int j=15;
   do
   {
      j-=2;
      cout<<j--<<endl;
   } while(j>14 );
return 0;
}
```

Upon execution, the program will display

A. 15
B. 14
C. 13
D. 12

27. Consider the given declaration of an array.

int sample_array[3][4];

This array can store up to

A. 7 integer values.
B. 12 integer values.
C. 16 integer values.
D. 20 integer values.

28. Consider the given statement.

strcpy(x,y);

Upon execution, the statement will copy the content of

A. variable x into variable y.
B. variable y into variable x.
C. variables x and y into another variable.
D. another variable into variables x and y.

29. Consider the following five values to be stored in an array named as **Values**.

    10 20 30 40 50

    The array declaration and initialisation to store these values is

    A. int Values[5] = {10, 20, 30, 40, 50};
    B. int Values{5} = [10, 20, 30, 40, 50];
    C. int Values[4] = {"10" "20" "30" "40" "50"};
    D. int Values{4} = ["10" "20" "30" "40" "50"];

30. The statement that shows the CORRECT use of arguments with a user-defined function named as **Test** is

    A. int Test(int x, int y);
    B. Test(int x, int y);
    C. int Test(x, y);
    D. Test(x, y);

31. Which of the following functions is always present in a C++ program?

    A. cin
    B. cout
    C. gets function
    D. main function

32. A variable declared in a user-defined function can be accessed

    A. within main function.
    B. only within that function.
    C. only outside that function.
    D. in any other function.

33. How many components are there in a function?

    A. Two
    B. Three
    C. Four
    D. Five

34. Which of the following descriptions is TRUE about a call by reference?

    A. Any change made to the parameter affects the argument
    B. Copies the address of an argument into the formal parameter
    C. Copies the actual value of an argument into the formal parameter
    D. Any change made to the parameter keeps the argument unchanged

Annual Examinations 2024 AKU-EB for Teaching & Learning only

35. Consider the given program.

```
#include <iostream>
using namespace std;
void T(int x)
{
cout << "Value of X is: " << x + 2 << endl;
}
void T(double  y)
{
cout << "Value of Y is: " << y + 7.25 << endl;
}
int main()
{
  T(20);
  T(50.75);
  return 0;
}
```

The function in the given program depicts the concept of

A. function overriding.
B. function inheritance.
C. function overloading.
D. function encapsulation.

36. In a C++ program, the keyword ‘inline’ in an inline function is used with

A. function call.
B. main function.
C. return function.
D. function declaration.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

37. Consider the given program.

```
#include <iostream>
using namespace std;
int main () {
  float a=17.8;
  float *b;
  b=&a;
  cout<<"a variable: "<<a<<endl;
  cout<<"b variable: "<<b<<endl;
  cout<<"*b variable: "<<*b<<endl;
  return 0;
}
```

Which of the following statements is TRUE about the program?

A. Variable a is a pointer variable.
B. Variable a stores the memory address of variable b.
C. Variable b stores the value of variable a.
D. Variable b is a pointer variable.

38. Consider the given code.

```
int j=20, k=34;
int *n=j, *m=&k;
n=m;
```

The CORRECT interpretation of this code is that

A. m points to j.
B. n points to k.
C. value of k is assigned to j.
D. value of j is assigned to k.

Annual Examinations 2024 AK I-EB for Teaching & Learning only

39. Consider the given program.

```
#include <iostream>
using namespace std;
int main()
{
   int a = 10;
   int b = 20;
   int *c;
   c = &a;
   a = 30;
   *c = 40;
   cout << "Value of a: "<< a <<endl;
   cout << "Value of b: " << b<< endl;
}
```

If the line *c = 40; is removed from this program, then the output will be

Value of a: 10 Value of b: 10	Value of a: 10 Value of b: 30
A	B
Value of a: 30 Value of b: 10	Value of a: 30 Value of b: 20
C	D

40. Two objects named as **num1** and **num2** are created that belong to a class named as **Calculate**. The code for this task is

A. num1:num2 Calculate;
B. Calculate num1:num2;
C. num1, num2 Calculate;
D. Calculate num1, num2;

41. In object-oriented programming (OOP), a key feature that allows to create more than one function in a program having the same name but different signatures is

A. function overriding.
B. function inheritance.
C. function overloading.
D. function encapsulation.

42. Which of the following is a CORRECT characteristic of a destructor?

A. It is used to release the memory from an object of a class.
B. It is used to give initial value to the object of a class.
C. It has the same name as the class name.
D. It can be overloaded.

APU-EB Annual Examinations 2024 for Teaching & Learning only

43. Consider the given program.

```
#include <iostream>
using namespace std;
class P
{
public:
   float ans;
   P()
   {
      ans = 3;
   }
   P(int a)
   {
     ans = a * a;
   }
   P(int b, int c)
   {
      ans = b + c;
   }
   void show()
   {
      cout<< ans<< endl;
   }
};
int main()
{
   P z1;
   P z2(7);
   P z3(10,15);
   z1.show();
   z2.show();
   z3.show();
   return 0;
}
```

AKU-EB Annual Examinations 2024 for Teaching & Learning only

The output of this program is

0 7 25	3 49 25
A	B
0 25 7	3 25 49
C	D

44. Consider the given program.

```
#include<iostream>
using namespace std;
class RectArea
{
  private:
  int a;
  public:
  int area(int l, int b)
  {
     a = l * b;
     return a;
  }
};
int main()
{
  RectArea A;
  cout << A.area(4,5);
  return 0;
}
```

The number of data members that can be accessed ONLY within class is

A. one.
B. two.
C. three.
D. four.

AKU-EB Annual Examinations 2024 for Teaching & Learning only

45. Consider the given program.

```
#include <iostream>
using namespace std;

class BankAccount {
public:
    string owner;
    double balance;

    void deposit(double amount) {
        balance += amount;
    }

    void withdraw(double amount) {
        balance -= amount;
    }
};

int main() {
    BankAccount account;
    account.owner = "Sameen";
    account.balance = 900.0;

    account.deposit(500.0);
    account.withdraw(200.0);
    cout << "Final balance for " << account.owner << ": $" << account.balance << endl;

    return 0;
}
```

AKU-EB Annual Examinations 2024 for Teaching & Learning only

If Sameen deposits $700 into his account and later withdraws $400, then the final balance will be

A. $300
B. $700
C. $1100
D. $1200

46. Consider the given program.

```
#include <iostream>
using namespace std;
class MyClass {
private:
   int privateVar=15;

public:
   MyClass() {
      privateVar = 10;
   }

   int getPrivateVar() {
      return privateVar;
   }
};
int main() {
   MyClass obj;

   int value = obj.getPrivateVar();
   cout << value << endl;

   return 0;
}
```

The output of this program is

A. 5
B. 10
C. 15
D. 25

AKU-EB Annual Examinations 2024 for Teaching & Learning only

47. Consider the given program.

```
#include <iostream>
using namespace std;

class B {
public:
   int value;

   B(int val) : value(val) {}
};

class D : public B {
public:
   int value;

   D(int val1, int val2) : B(val1), value(val2) {}
};

int main() {
   D obj(10, 30);
   cout << obj.value << endl;
   return 0;
}
```

The output of this program is

A. 10
B. 20
C. 30
D. 40

AKU-EB Annual Examinations 2024 for Teaching & Learning only

48. Consider the given program.

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
   ifstream file("sample.txt");
   char ch;

   while (!file.eof()) {
      file.get(ch);
      cout << ch;
   }

   return 0;
}
```

The purpose of while loop in the given code is that it

A. writes characters to a file.
B. searches for a specific character in the file.
C. counts the number of characters in the file.
D. reads characters from a file and displays them.

49. What is the recommended practice for closing a file in C++?

A. Deleting the file
B. Using the fclose() function
C. Using the file.close() method
D. Automatically closes when the program exits

50. In a C++ program that reads a file into strings, the significance of the eof() function is that it

A. opens the file for reading.
B. counts the number of lines in the file.
C. checks if the file exists in the filesystem.
D. determines if the end of the file has been reached.

AKU-EB Annual Examinations 2024 For Teaching & Learning only

END OF PAPER

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2024
for Teaching & Learning only